ERPNext API Integration Guide

Complete Technical Workflow: Material Request (PR) to Purchase Order (PO)

Document Purpose	End-to-End REST API Workflow & Technical Reference	**Target System**	ERPNext v13 / v14 / v15
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1. Executive Summary & Root Cause Analysis

This document details the exact REST API lifecycle required to create a **Material Request (Purchase Request)**, submit it, convert it into a **Purchase Order (PO)**, and submit the PO in ERPNext so that the original Material Request successfully transitions to the `ORDERED` state (green signal).

> **CRITICAL ARCHITECTURAL REQUIREMENT: Line-Item Mapping**
>
> In ERPNext, creating a Purchase Order with a top-level `material_request` ID is **insufficient** to update the parent document's status. ERPNext tracks fulfillment at the child-item level.
>
> Each row in the Purchase Order's `items` array **MUST** explicitly contain both:
>
> - `material_request`: The parent Material Request document ID (e.g., `MAT-MR-2026-00008`).
> - `material_request_item`: The unique child row ID (hash key) from the Material Request (e.g., `hhq07urcug`).
>
> If `material_request_item` is missing, ERPNext creates an unlinked Purchase Order, leaving the Material Request permanently trapped in **Pending** state.

2. Complete Step-by-Step API Execution Flow

Step 1: Create a Draft Material Request (PR)

Initiate the procurement process by creating a draft Material Request document of purpose `Purchase`.

`POST` `/api/resource/Material Request`

Request Headers

```
Authorization: token <api_key>:<api_secret>
Content-Type: application/json
```

Request Payload

```
{
  "material_request_type": "Purchase",
  "schedule_date": "2026-09-01",
  "company": "Maahi's Foudation",
  "items": [
    {
      "item_code": "ITEM-001",
      "qty": 10,
      "schedule_date": "2026-09-01",
      "warehouse": "Stores - MF"
    }
  ]
}
```

Expected Response Highlights

```
{
  "data": {
    "name": "MAT-MR-2026-00008",
    "docstatus": 0,
    "status": "Draft",
    "items": [
      {
        "name": "hhq07urcug",  // <-- RECORD THIS CHILD ROW ID
        "item_code": "ITEM-001",
        "qty": 10
      }
    ]
  }
}
```

Step 2: Submit the Material Request

Submitting the Material Request locks it and sets its status to **Pending**.

`PUT` `/api/resource/Material Request/MAT-MR-2026-00008`

Request Payload

```
{
  "docstatus": 1
}
```

Expected Status

Document `MAT-MR-2026-00008` is now in `PENDING` status.

Step 3: Query Material Request Row Metadata (If needed)

If the child row ID was not stored during Step 1, perform a GET call to retrieve the item table metadata.

`GET` `/api/resource/Material Request/MAT-MR-2026-00008`

Locate the `name` property inside the `items` array (e.g., `"name": "hhq07urcug"`).

Step 4: Create Draft Purchase Order (PO) with Exact Row Linking

Construct the draft Purchase Order. Ensure both `material_request` and `material_request_item` are supplied in the child table.

`POST` `/api/resource/Purchase Order`

Request Payload

```
{
  "supplier": "Test Supplier API",
  "schedule_date": "2026-09-01",
  "company": "Maahi's Foudation",
  "items": [
    {
      "item_code": "ITEM-001",
      "qty": 10,
      "rate": 100,
      "warehouse": "Stores - MF",
      "material_request": "MAT-MR-2026-00008",
      "material_request_item": "hhq07urcug"  // <-- MANDATORY LINE LINK
    }
  ]
}
```

Expected Response Highlights

```
{
  "data": {
    "name": "PUR-ORD-2026-00010",
    "docstatus": 0,
    "status": "Draft"
  }
}
```

Step 5: Submit the Purchase Order

Submitting the Purchase Order triggers the internal ERPNext update hook that updates the status of the linked Material Request.

`PUT` `/api/resource/Purchase Order/PUR-ORD-2026-00010`

Request Payload

```
{
  "docstatus": 1
}
```

Expected Status

Purchase Order `PUR-ORD-2026-00010` transitions to `To Receive and Bill`. The Material Request `MAT-MR-2026-00008` automatically flips to `ORDERED`.

3. Alternative Method: Native Mapper Endpoint

To completely bypass manual child-table row key management, ERPNext provides a built-in Python controller method that automatically maps all document fields and row IDs:

`POST` `/api/method/erpnext.stock.doctype.material_request.material_request.make_purchase_order`

Request Payload

```
{
  "source_name": "MAT-MR-2026-00008"
}
```

> **Mapper Endpoint Workflow**
>
> 1. Send the above request; ERPNext returns a pre-populated, perfectly linked Purchase Order dictionary in `response.message`.
> 2. Inject your `supplier` name into the dictionary.
> 3. `POST` that dictionary directly to `/api/resource/Purchase Order`.
> 4. `PUT` to submit the Purchase Order (`{"docstatus": 1}`).

4. Troubleshooting & UI Caching Notes

- **Browser Cache Issues:** ERPNext's frontend desktop uses heavy client-side caching. After completing API calls via Postman, open the Material Request list view and perform a hard refresh (`Ctrl + Shift + R` or `Cmd + Shift + R`) to reflect the green **Ordered** status.
- **Partial Quantities:** If a Material Request requests 10 units but the PO is submitted for only 5 units, the Material Request status will be set to **Partially Ordered** instead of **Ordered**.

- **Active Workflows:** If a custom workflow is attached to the Material Request DocType, workflow state logic takes priority over ERPNext's standard automated status triggers.